



Comparative GIS-Based Ensemble Machine Learning for Groundwater Potential Mapping and Hydrogeological Controls in the Arid Stampriet Transboundary Aquifer System, Namibia

Asteria Lea Mwetulundila^{1,3} · Ernestine Atangana^{2,3}

Received: 22 November 2025 / Revised: 12 February 2026 / Accepted: 3 March 2026 / Published online: 24 March 2026
© The Author(s) 2026

Abstract

Groundwater potential mapping is increasingly used to support well siting and resource planning in arid regions, particularly where hydrogeological data are scarce. This study presents the first GIS based assessment of groundwater potential for the Namibian section of the Stampriet Transboundary Aquifer System (STAS). It applies comparative ensemble machine learning models, namely Boosted Regression Trees (BRT) and Stacked Ensemble Random Forest (SERF). These were selected for their ability to capture nonlinear hydrogeological interactions and maintain predictive robustness under data-limited and heterogeneous conditions. The workflow, including model parameterisation and input datasets, is fully documented in the methods section to support reproducibility in principle. Predictor variables representing hydrogeomorphological, geological, and climatic controls, including drainage and lineament proximity and density, depth to bedrock, terrain attributes, rainfall, bulk density, and surface wetness, were integrated within BRT and SERF. Model performance was evaluated using five-fold spatial block cross-validation to reduce bias arising from spatial autocorrelation and assessed using receiver operating curve–area under curve (ROC–AUC) and frequency ratio analyses. Although both models achieved high predictive accuracy ($AUC > 0.90$), results remain influenced by data scarcity and hydrogeological heterogeneity and should therefore be interpreted as probabilistic indicators rather than deterministic predictions. Both models produced coherent groundwater potential patterns, consistently delineating a north-central to north-western corridor of elevated groundwater potential extending from Stampriet towards Gross Ums and Onderombapa. BRT demonstrated higher predictive skill ($AUC = 0.958$) than SERF ($AUC = 0.915$), indicating stronger representation of nonlinear hydrogeological interactions. Drainage proximity, net recharge, rainfall, and depth to bedrock emerged as the dominant predictors. These demonstrate the value of focused recharge along ephemeral drainage pathways, combined with subsurface storage constraints in this recharge-limited, multi-layered aquifer system. High to very high groundwater potential zones occupy a limited proportion of the basin (7.08% for BRT; 12.04% for SERF). This provides a practical basis for prioritising exploration, siting efficiency of new production boreholes, and supporting groundwater licensing, drought planning, and transboundary aquifer governance in the STAS.

✉ Asteria Lea Mwetulundila
mwetulundilaal@gmail.com

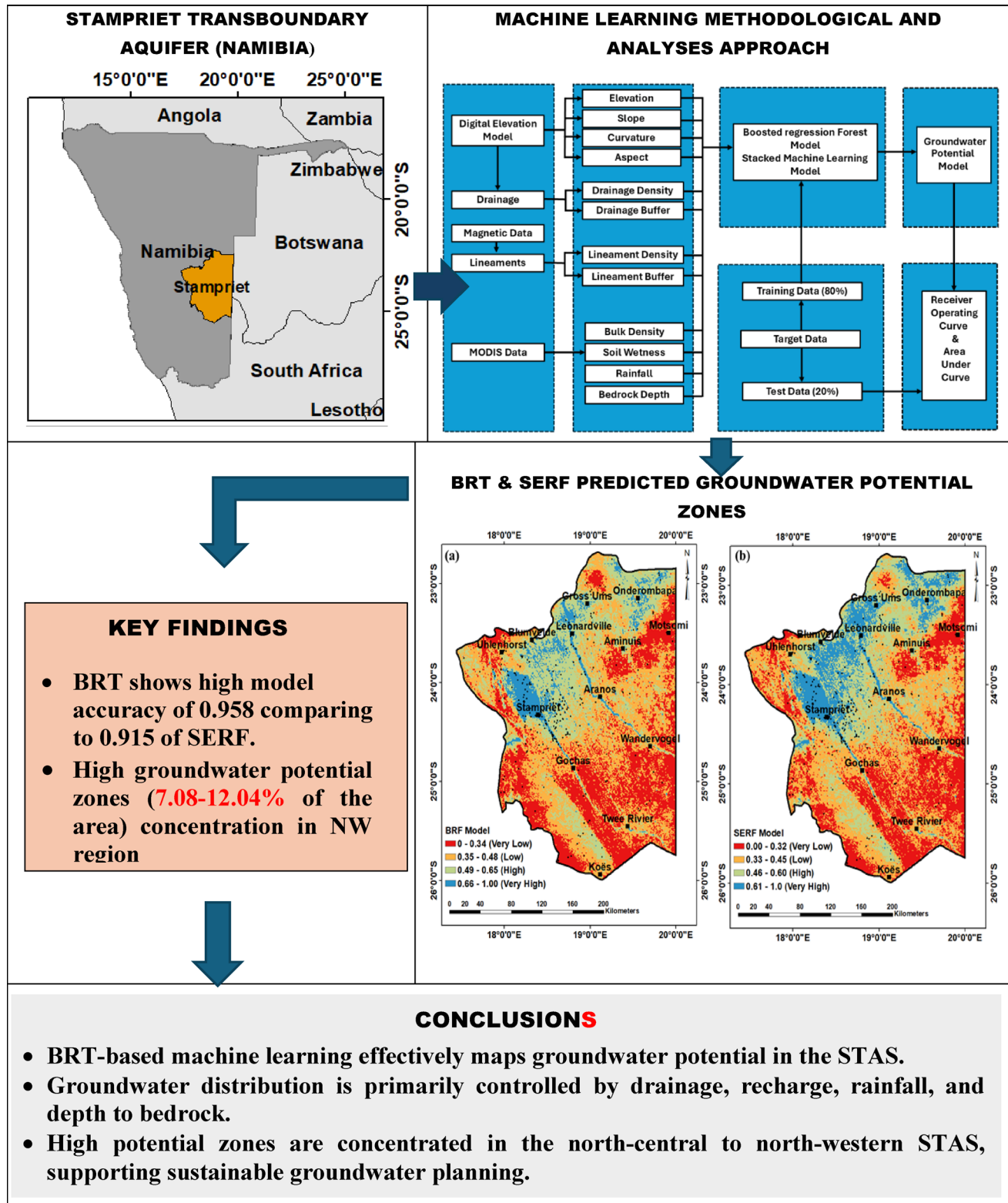
✉ Ernestine Atangana
AtanganaE@ufs.ac.za

¹ Institute for Groundwater Studies, Bloemfontein, South Africa

² Centre for Environmental Management, Bloemfontein, South Africa

³ Faculty of Natural and Agricultural Sciences, University of the Free State, 339, Bloemfontein 9300, Republic of South Africa

Graphical Abstract



This graphical abstract presents a concise visual representation of groundwater potential mapping within the Namibian sector of the Stampriet Transboundary Aquifer System (STAS). It depicts the study area, along with the compiled geospatial and hydrogeological datasets, and outlines the machine-learning workflow employed to delineate groundwater potential zones. The framework assesses Boosted Regression Trees (BRT) and Stacked Ensemble Random Forest (SERF) models,

emphasising the superior predictive performance of BRT, as evidenced by high area under curve (AUC) values. The findings highlighted drainage networks, net recharge, rainfall, and depth to bedrock as the critical factors influencing groundwater occurrence. High potential zones were predominantly mapped in the north-central to north-western portions of the STAS, accounting for approximately 7.08–12.04% of the total area. These spatially explicit outputs offer actionable insights for targeted groundwater development, aquifer protection, and sustainable resource management in both arid and semi-arid environments. The study also addressed methodological limitations and suggests future improvements, including the integration of subsurface data, robust spatial validation and reproducible workflows. It also included a thorough consideration of the multi-layered aquifer system, strengthening the study's contribution to data-driven, sustainable groundwater management.

Highlights

- Groundwater potential mapped in the Namibian sector of the Stampriet Transboundary Aquifer System using GIS–ML models.
- Boosted Regression Trees (BRT) and Stacked Ensemble Random Forest (SERF) trained using borehole data and spatial block cross-validation.
- BRT outperformed SERF in groundwater potential prediction based on receiver operating curve–area under curve (ROC–AUC) metrics.
- Drainage proximity, recharge, rainfall, and depth to bedrock identified as key groundwater controls.
- High groundwater potential zones are limited and concentrated in the north-central to north-western parts of the basin, supporting targeted groundwater management.

Keywords Stampriet transboundary aquifer system · Groundwater potential mapping · Stacked ensemble random forest · Boosted regression trees · Spatial cross-validation · Machine learning models · Namibia

1 Introduction

Water scarcity is intensifying globally as a result of over-exploitation, pollution, and climate variability, with billions of people experiencing seasonal or chronic water stress (Golovina and Grebneva 2021; Mekonnen and Hoekstra 2016; UN-Water, 2025). In Sub-Saharan Africa, these pressures are exacerbated by limited and unreliable surface water supplies, increasing dependence on groundwater as a comparatively resilient source for domestic, agricultural, and industrial uses (Calow et al. 2010; Mapani et al. 2023). Groundwater occurrence is spatially heterogeneous in arid and semi-arid regions, requiring methods that can identify favourable hydrogeological conditions at large spatial scales.

Traditionally, groundwater exploration has relied on field-based geological mapping and geophysical surveys (Adeoye et al. 2025; Odoh et al. 2024; Rodriguez and Fero-lin 2024). These methods can be highly accurate at a local scale; however, they are often costly, time-consuming, spatially limited, and require specialised expertise. This limitation is particularly pronounced in resource-constrained settings (Ahaneku et al. 2025; Alao and Abubakar 2025). Consequently, the integration of Remote Sensing and Geographic Information Systems (RS–GIS) has become central to regional-scale groundwater prospecting. These approaches enable the synthesis of multisource spatial datasets and surface proxies linked to groundwater occurrence

such as lithology, geological structures, drainage networks, and terrain attributes into groundwater potential maps (Corner and Durrheim 2018; Yusuf et al. 2025). RS–GIS groundwater mapping has evolved from classical multicriteria approaches, such as AHP and fuzzy logic, and bivariate statistical methods including frequency ratio and weights of evidence. Recent studies increasingly apply machine-learning techniques to capture nonlinear relationships and reduce subjectivity in factor weighting (Bamal et al. 2024; Haghiri et al. 2024; Pappaka et al. 2025; Pawar and Sharma 2025; Saleh and Rasel 2024). Despite their advantages, machine learning models are sensitive to the availability, quality, and spatial representativeness of training and validation data. In arid regions, this dependence on borehole yields, groundwater levels, hydrochemical, and geophysical data often limits reliability in data scarce settings.

Namibia exemplifies such a context, being predominantly arid with low rainfall, extremely high evapotranspiration, and limited perennial surface water. As a result, groundwater constitutes the primary source of water for livelihoods, economic activities, and ecosystem support (Mapani et al. 2023; Nuunyango 2019; Shikangalah 2020). Limited recharge, variable borehole yields, groundwater quality concerns, and recurrent droughts compound these challenges (Li et al. 2018; Shikangalah 2020; Uugulu and Wanke 2020). In this context, the STAS, shared by Namibia, Botswana, and South Africa, is a strategically critical groundwater reserve

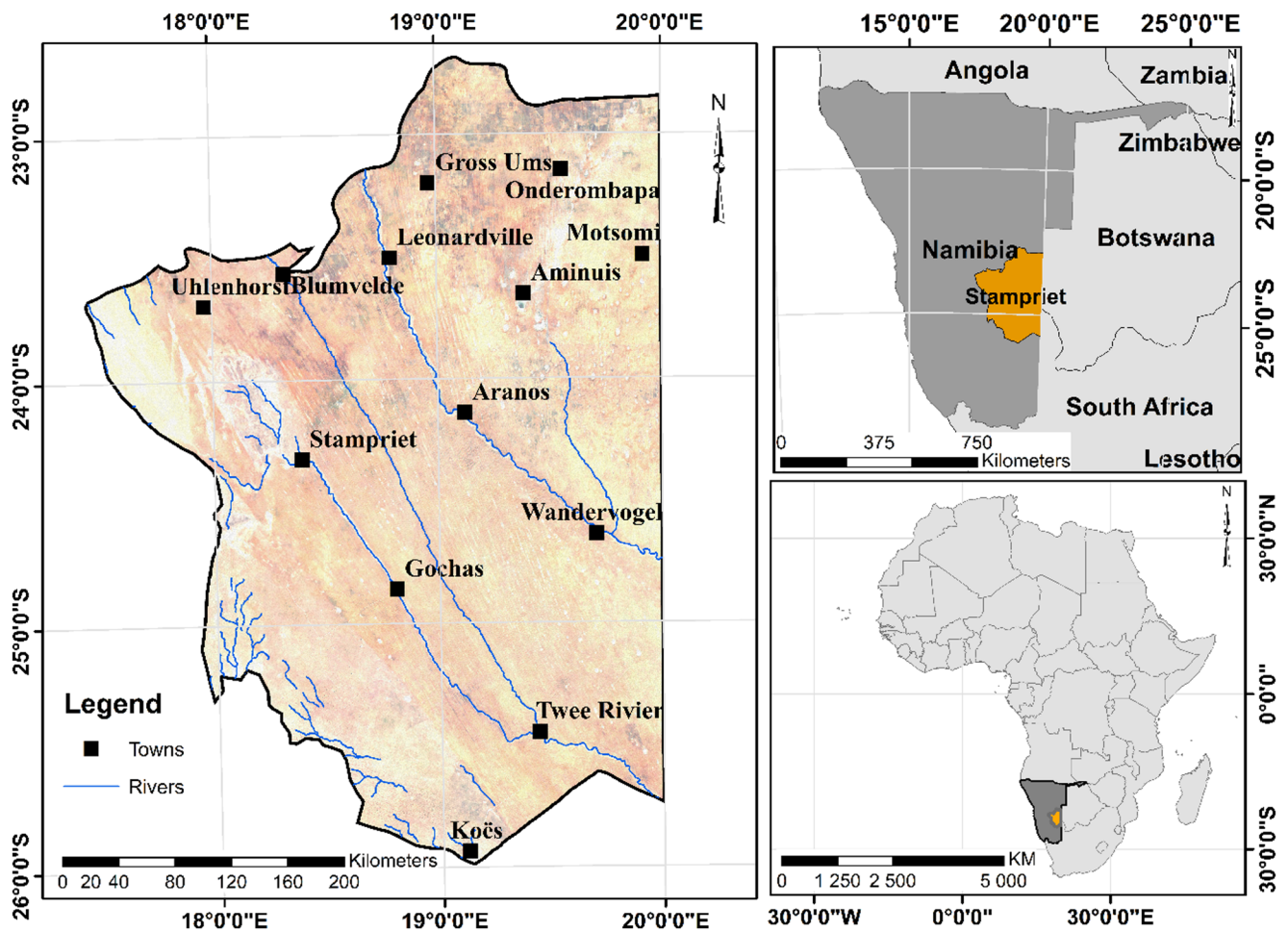


Fig. 1 Location of the Stampriet Transboundary Aquifer on the Namibian side

where surface water is limited to ephemeral rivers (Lohe et al. 2020; Miller 2013, 2024; Miller et al. 2018).

The Stampriet Transboundary Aquifer System (STAS) is a complex, multi-layered sedimentary aquifer system, where shallow unconfined units overlie deeper confined aquifers, resulting in intricate recharge–storage–flow interactions (Cihan et al. 2014; Miller et al. 2018). Despite its regional importance, the groundwater potential of the STAS remains insufficiently explored. Only a limited number of studies have been conducted, most of which relied on numerical groundwater modelling and satellite-based interpretations derived from GRACE observations (Kinoti et al. 2024). GIS-based machine-learning models are valuable for simulating groundwater processes. However, in the multi-layered STAS, the coarse spatial resolution of available data limits representation of local hydrogeological variability, reducing predictive reliability (Kinoti et al. 2024). Similarly, GRACE observations provide insights into large-scale terrestrial water storage, but their coarse resolution prevents detailed local-scale analysis, limiting suitability for site-specific management (Ali et al. 2021; Kinoti et al. 2024).

More broadly, RS–GIS groundwater studies in arid regions have often focused on conceptual or deterministic frameworks, such as morphometric analysis and multicriteria evaluation/Analytical Hierarchy Process (AHP) overlays. These approaches yield generalized suitability maps useful for reconnaissance assessments but are less effective for operational applications, including borehole siting, abstraction licensing, and protection zoning (Khan and ElKashouty 2023; Khan et al. 2022a, b, 2023). Many studies also have limited validation, often relying on well-location coincidence or broad hydrochemical indicators (e.g., TDS, EC, and nitrate). They rarely adopt spatially robust validation strategies or explicitly consider predictive uncertainty, reducing confidence in model outputs for regulatory use and transboundary water governance (Khan et al. 2022a, b, 2023).

To address these challenges, recent studies increasingly advocate integrating GIS with advanced machine-learning and ensemble modelling techniques to improve groundwater assessments in data-constrained environments. Such approaches can enhance predictive accuracy, reduce bias,

and better represent complex hydrogeological processes (Naghibi et al. 2016; Sachdeva and Kumar 2021). Despite growing global application, GIS-based machine learning methods remain underused in complex transboundary aquifers, particularly in hyper-arid regions where reliable groundwater information is most urgently needed (Calow et al. 2010; Golovina and Grebneva 2021). Moreover, only a limited number of studies have applied these approaches through ensemble learning techniques, such as Boosted Regression Trees (BRT) and Stacked Ensemble Random Forest (SERF), even though ensemble methods reduce prediction error and improve robustness under environmental uncertainty and data scarcity (Halder et al. 2024; Naghibi et al. 2016). These approaches have limitations, including higher computational demand, reduced interpretability, and sensitivity to hyperparameter selection. Nevertheless, they consistently outperform conventional statistical and single machine learning models in nonlinear groundwater and hydrological applications (Naghibi et al. 2016; Xu et al. 2024).

In response to these methodological and contextual gaps, this study advances groundwater assessment in the STAS by implementing a GIS-based ensemble machine-learning framework tailored to data-limited, arid, and transboundary aquifers. By integrating hydrogeological, climatic, and

remotely sensed datasets, the study provides a systematic evaluation of groundwater potential under hyper-arid conditions and structural complexity. The comparative application of BRT and SERF enables robust assessment of nonlinear process interactions, which are poorly captured by conventional numerical or deterministic approaches. Although applied to the STAS, the methodological framework and validation strategy are broadly applicable to groundwater potential assessment in arid, data-limited, and transboundary aquifer systems globally.

A key methodological contribution of this study is the adoption of spatially explicit validation strategies. This includes spatial block cross-validation combined with complementary performance metrics (e.g. Receiver Operating Curve (ROC–AUC) and Precision Recall–Area under Curve (PR–AUC) to reduce over-optimistic estimates caused by spatial autocorrelation. Interpretable diagnostics, such as permutation importance and SHapley Additive exPlanations (SHAP) analysis, enable transparent identification of the dominant geological and hydrological controls on groundwater occurrence in the STAS. These approaches enhance both the scientific credibility and operational interpretability of the resulting groundwater potential maps. Beyond methodological advancement, this study translates model outputs

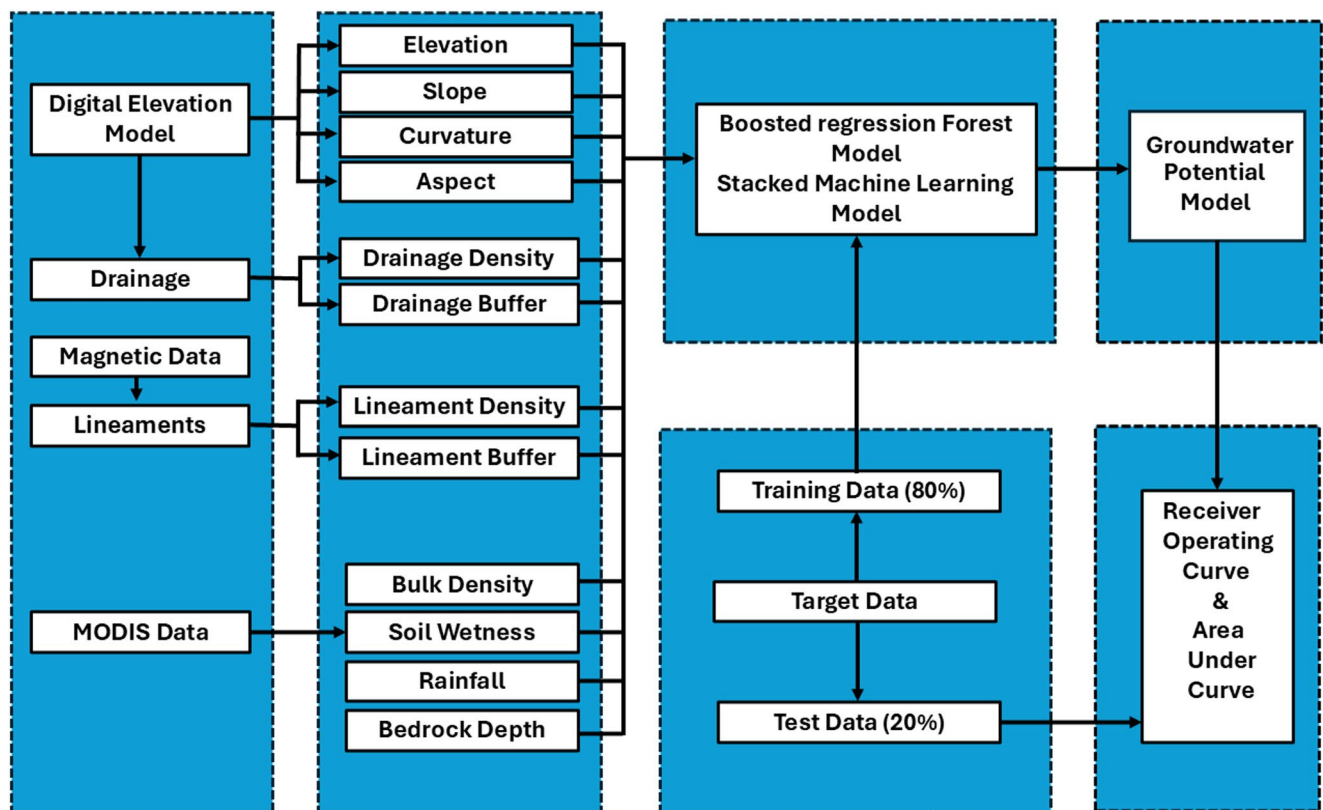


Fig. 2 Presents the methodological workflow used to develop the groundwater predictive model for the Stampriet Basin. It shows the key processing steps for data preparation and model development

into actionable spatial information relevant to groundwater governance and management.

The delineation of discrete high potential and recharge zones provides direct support for borehole siting, abstraction licensing, drought preparedness, and protection zoning within the Namibian sector of the STAS. Importantly, these outputs address a critical governance gap in transboundary groundwater management, where uneven dependence on the aquifer and increasing anthropogenic pressures in upstream recharge areas necessitate coordinated, evidence-based decision-making. By explicitly linking advanced groundwater potential mapping to governance needs, this research contributes a transferable framework for sustainable groundwater management in arid transboundary aquifer systems. The approach supports collaborative planning between Namibia, Botswana, and South Africa. It provides spatially defensible evidence to safeguard groundwater quantity and quality amid climate variability, development pressures, and long-term water insecurity (Nijsten et al. 2018; ORASECOM 2016).

2 Description of the Study Area

2.1 Geographical Location and Climatic Description

The STAS is situated along the eastern axis of Namibia, extending into the Republic of South Africa and Botswana between longitudes E17°22'00" to E20°05'00" and latitudes S22°17'00" to S26°08'00" (Fig. 1). The Namibian portion of STAS covers approximately 63,254 km². The terrain

slopes gently from northwest to southeast, ranging in elevation from 1,500 m to 900 m above mean sea level, and is predominantly flat. The STAS comprises a multi-layered sedimentary aquifer system, with shallow unconfined Kalahari units overlying deeper confined Auob and Nossob aquifers. These formations control recharge, storage, and flow dynamics (Cihan et al. 2014). Groundwater flow generally follows the regional hydraulic gradient from northwest toward southeast, consistent with the dip of the underlying geological formations (Miller 2008). However, Kinoti et al. (2024) reported that groundwater also flows eastward into Botswana before diverging toward the southeast, south, and southwest along the lower Nossob River. This reflects a bifurcation in regional groundwater flow paths. Recharge occurs primarily through ephemeral rivers and localized rainfall infiltration, with recharge rates highly variable due to aridity and spatial heterogeneity (Lohe et al. 2020; Miller et al. 2018).

Groundwater discharge is controlled by abstraction, evapotranspiration, and occasional spring emergence. The climate is hyper-arid to semi-arid, with annual rainfall ranging from 150 mm in the west to 400 mm in the northeast (Mendelsohn et al. 2002). Temperatures average 22–28 °C, contributing to high evaporation rates and limited surface water availability (Mapani et al. 2023; Ugulu and Wanke 2020). The STAS supports critical water supply for domestic, agricultural, and industrial use across Namibia, Botswana, and South Africa, making accurate groundwater assessment essential for sustainable management and transboundary governance.

Table 1 Exploration datasets used for predictive mapping of groundwater, including data source, data processing techniques applied and the software used

Variable	Source	Processing Technique	Software	Resolution
Elevation	ASTER Global Digital Elevation Map (GDEM 003) (Jet Propulsion Laboratory 2004)	Fill algorithm	ArcMap 10.3	30 m per pixel (1 arc-second Resolution)
Slope	ASTER DEM (GDEM 003)	Slope tool	ArcMap 10.3	1 × 1 km
Profile Curvature	ASTER DEM (GDEM 003)	Curvature tool	ArcMap 10.3	1 × 1 km
Aspect	ASTER DEM (GDEM 003)	aspect tool	ArcMap 10.3	1 × 1 km
Drainage Buffer	DEM-derived drainage network	Euclidean Distance algorithm	ArcMap 10.3	1 × 1 km
Drainage Density	DEM-derived drainage network	Kernel density algorithm	ArcMap 10.3	1 × 1 km
Lineament Buffer	Aeromagnetic data (Namibian Geological Survey)	Euclidean distance algorithm	ArcMap 10.3	1 × 1 km
Lineament Density	Aeromagnetic data	Kernel density tool algorithm	ArcMap 10.3	1 × 1 km
Bedrock Depth	International Soil Reference and Information Centre	Subsetting	ArcMap 10.3	250-meter spatial resolution globally
Bulk Density	International Soil Reference and Information Centre	Data averaging	ArcMap 10.3	250 m
Rainfall	CHRS iRain (2025)	Resampling, sub setting	ArcMap 10.3	~0.04° resolution
Wetness	MODIS Spectral Réflectance (NASA; https://earthdata.nasa.gov)	Band ratio	Envi 5.6	500 m

Sources: Maqsoom et al. 2022; Olaleye 2024)

2.2 Geology and Hydrogeology

The groundwater potential of the STAS is governed by its geological and hydrogeological framework. It is characterised by a south-easterly dipping Karoo sedimentary succession overlain by unconsolidated Kalahari Group sand and calcrete (Kinoti et al. 2024). Permeability contrasts between highly porous Kalahari sediments, fractured sandstones of the Auob and Nossob aquifers, and low-permeability Dwyka and Prince Albert formations create a vertically layered flow system with distinct recharge, storage, and transmission characteristics (Miller 2008). Recharge occurs primarily along ephemeral river channels and dune–interdune corridors, where focused infiltration reaches shallow aquifers. Diffuse recharge to deeper Auob and Nossob units is limited by regional confinement. Structural features, including faults and lineaments, locally enhance groundwater potential by increasing secondary porosity and transmissivity. Clay-rich fault gouge can impede flow in some areas, creating hydraulic barriers (Lohe et al. 2020).

The STAS comprises four sedimentary aquifers, arranged from bottom to top: Dwyka, Nossob, Auob, and Kalahari (Kinoti et al. 2024; Miller 2008, 2024; Tredoux et al. 2002). The Dwyka Group is largely impermeable in Namibia, confining the Nossob aquifer. Both the Nossob and Auob aquifers are artesian, but the Auob is gradually losing its artesian characteristics due to declining groundwater levels from over-abstraction and limited recharge (Miller 2013, 2024). Hydraulic connectivity between the Auob and Kalahari aquifers allows sub-artesian conditions in parts of the basin. The Auob and Kalahari aquifers are the primary groundwater sources. Groundwater quality is generally lower in the Kalahari aquifer, but its shallow depth facilitates exploitation and reduces drilling costs (Kinoti et al. 2024; Lohe et al. 2020; Miller 2024).

Aquifer thickness varies spatially, reflecting stratigraphy and erosional history. Erosion of the Rietmond Formation in central and southern areas has created direct hydraulic connection between the Auob and Kalahari aquifers, contrasting earlier interpretations suggesting minimal connectivity (JICA 2002; Stone and Edmunds 2012; Tredoux et al. 2002). Geological structures significantly influence groundwater flow and recharge. Faults introduce hydraulic anisotropy: some restrict lateral flow, preserving palaeowaters, while others facilitate vertical leakage, aquifer connectivity, and potential cross-contamination (Corner and Durrheim 2018; Lohe et al. 2020; Miller 2024).

The multi-layered STAS is highly sensitive to anthropogenic activities. Approximately 33 exclusive prospecting licenses (EPLs) for uranium, covering approximately 53% of the Namibian part of the basin, have been issued (Miller 2024). The uranium deposits in the Auob Aquifer

are largely low-grade and amenable to acid in-situ leaching (ISL), which allows uranium recovery without excavation (Li and Yao 2024; Seredkin et al. 2016). The operational efficiency of ISL critically depends on sufficient water to maintain a deep cone of depression in the saturated, confined aquifer (Mudd 2022). This cone confines the leaching solution, optimizes uranium recovery, and prevents uncontrolled migration of chemicals. Inadequate water availability could compromise cone formation, reduce leaching efficiency, degrade groundwater quality, and lower water levels in surrounding boreholes (Mwetulundila and Atangana 2025). Given the limited natural recharge and high evaporation rates in the region, ISL could exacerbate water scarcity, particularly in high potential recharge zones of the STAS. Therefore, identifying and delineating these zones is therefore essential to support sustainable groundwater management amid increasing anthropogenic pressures.

3 Materials and Methods

3.1 Data collection and Processing

Accurate groundwater prediction requires integrating diverse geospatial and hydroclimatic datasets to capture the complex controls on aquifer occurrence. The following workflow details the data sources, processing steps, and modelling procedures used for the Stampriet Basin (Fig. 2).

Digital Elevation Model (DEM), aeromagnetic, and MODIS data were processed to extract relevant topographic, structural, soil, and climatic features. Predictor variables included elevation, slope, drainage density, lineament density, soil wetness, and rainfall, served as inputs into BRT and SERF models. Models were trained with 80% of observed groundwater data and tested on the remaining 20%. While, model performance was evaluated using Receiver Operating Curve (ROC) and Area under the Curve (AUC) metrics. The resulting models produced a spatially explicit groundwater potential map to guide exploration and resource management.

3.2 Data Source and Processing

Target and exploratory datasets, which enabled the development of groundwater predictive models for the STAS were chosen. The attributes of exploration datasets used for predictive mapping of groundwater is given in Table 1. The datasets provided comprehensive spatial and attribute information, which supported accurate identification of groundwater potential zones. The target dataset contained positive and negative samples, and were assigned values of 1 and 0, respectively. Positive samples, representing borehole

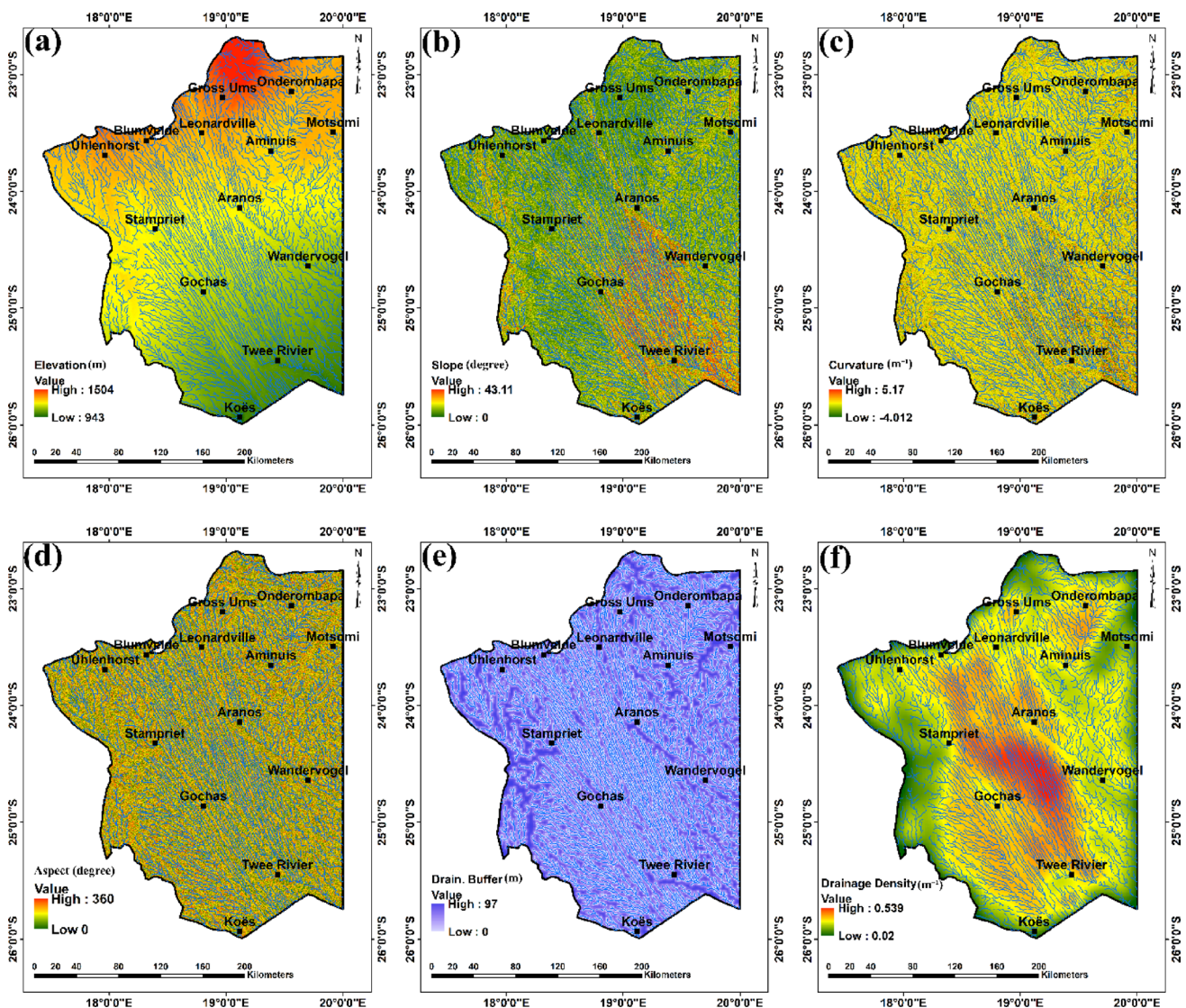


Fig. 3 Geomorphological and hydrological data for groundwater predictive mapping in the Stampriet area (a) elevation, (b) slope, (c) curvature, (d) aspect, (e) drainage buffer, (f) drainage density

locations with yields equal to or above Namibia's sustainable threshold, originated from the Ministry of Agriculture, Fisheries, Water and Land Reform's GROWAS Database. Negative samples, matching the count of positive samples, consisted of randomly generated points. The exploratory data employed for groundwater predictive modelling included multi-source data derived from environmental, geological, hydrological, and topographical factors, which collectively informed the identification of groundwater potential zones. These datasets included elevation, slope, curvature, aspect, drainage buffer, drainage density, lineament buffer, lineament density, bedrock depth, bulk density, and rainfall. The Jet Propulsion Laboratory at the University of California (2004), provided elevation data as an ASTER Global Digital Elevation Model Version 3 (GDEM 003). The fill algorithm in ArcMap enabled the elimination of minor

imperfections in the DEM data. The surface algorithm in ArcMap 10.3 extracted slope data from the DEM, quantifying terrain variations as slope angles in degrees. The DEM data served as the primary source for extracting profile curvature and aspect datasets using the surface algorithm in ArcMap 10.3. The ArcMap 10.3 hydrology tool extracted the study area's drainage network from the DEM data. The Euclidean distance algorithm calculated distances from the drainages in metres to generate drainage buffer data. Structural parameters like lineament density and lineament buffers were developed from high resolution aeromagnetic data from the Geological Survey of Namibia. The magnetic data was transformed using the Reduced to Equator Filter, followed by the Tilt Derivative and Horizontal Tilt Derivative algorithms. Structural features were obtained through manual digitisation using ArcMap 10.3 software. The Euclidean

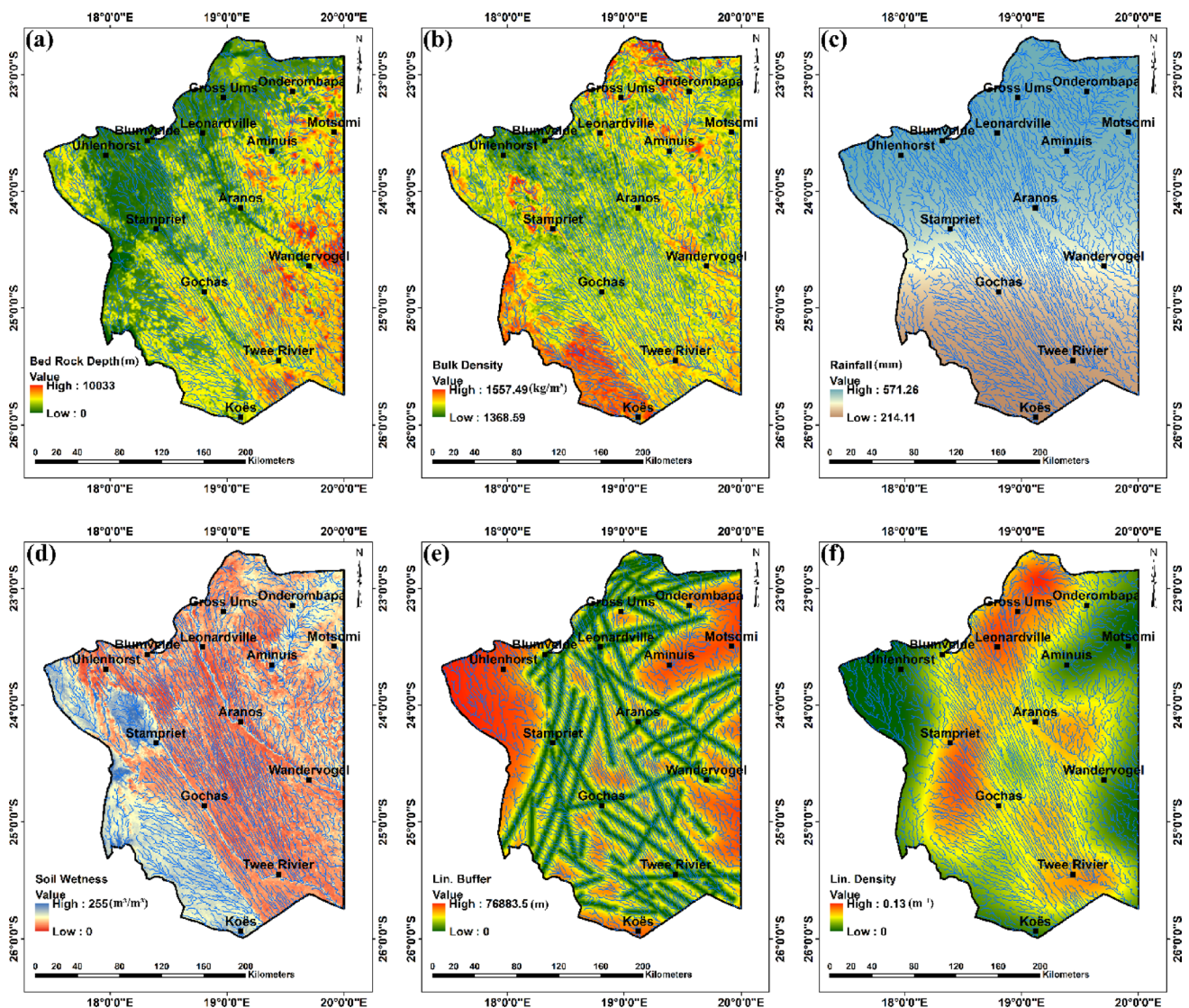


Fig. 4 Geological and environmental data for groundwater predictive mapping in Stampriet area (a) bedrock depth, (b) bulk density, (c) rainfall, (d) soil wetness, (e) lineament buffer, (f) lineament density

distance and kernel density algorithms were used to create lineament buffer and density datasets for the study.

The depth to bedrock data and bulk density datasets were downloaded from a global dataset provided by the International Soil Reference and Information Centre (ISRIC 2025). The soil bulk density data at six standard depths was reprocessed by averaging the bulk density based on the given depths. Rainfall data was downloaded from the Centre for Hydrometry and Remote Sensing (CHRS iRain 2025), using the ArcMap tool for raster calculations. The rainfall data have a spatial resolution of $0.04^\circ \times 0.04^\circ$, including an average distribution for multiple years. Mean rainfall data was generated using the ArcMap calculator, and resampling was performed to enhance the spatial resolution. Wetness data were derived from MODIS spectral reflectance data from the National Aeronautics and Space Administration

(NASA). A band ratio image of 4/2 was computed using the ENVI software.

Prior to model training, predictor variables were screened for multicollinearity using pairwise correlation analysis (Shrestha 2020). Highly correlated variables were removed or retained based on hydrogeological relevance to reduce redundancy and improve model stability.

3.3 Data and Sampling Design

Sampling design and class balance positive samples were obtained from boreholes with sustainable yield $\geq$ threshold [L/S] from GROWAS. Negative (pseudo-balance) were generated by stratified random sampling at ≥ 3 km from any positive, outside mapped wellhead protection areas, and masked from surface water features to reduce label noise.

Class balance was enforced ($P=N$) and robustness checked at 1:2 and 2:1 ratios.

3.4 Spatial Cross Validation

To mitigate spatial autocorrelation bias, we used 5-fold spatial block cross-validation (≈ 25 – 50 km blocks), rotating held-out blocks across the basin. All pre-processing (normalisation, feature selection) was nested within each fold to avoid data leakage.

4 Exploration Data Characteristics and Recognition Factors

Exploration datasets displayed diverse spatial characteristics, which indicated the presence of groundwater in the study area (Fig. 3). Elevation in the study area ranged from 943 to 1,504 m above sea level, with highlands concentrated in the northern part of STAS. Elevation influences groundwater recharge, with higher elevations functioning as recharge zones and lower elevations serving as discharge zones. Gravity directs groundwater toward these lower regions, increasing the potential water supply at lower altitudes. The slope data varies from 0 to 43.0 degrees with steep slopes occurring predominantly in the central to the southern axis of the study location. Slope has an important effect on runoff and infiltration patterns. Steeper gradients lead to heightened surface runoff and reduced infiltration, which consequently impedes groundwater recharge. In contrast, gentle or flat slopes promote better infiltration, thus enhancing groundwater recharge and storage capacity. Curvature in the STAS area ranged from -4.02 to 5.17 m^{-1} , with both concave and convex forms widely distributed; convex slopes were most prominent in the central-eastern and south-eastern sections. Curvature (expressed in m^{-1}) serves as an important indicator of topographic form and influences how water is retained or drained across the landscape. Convex surfaces, defined by positive curvature, facilitate water runoff; while concave surfaces, marked by negative curvature, tend to collect and retain water. These concave areas are more effective at capturing and infiltrating water, in so doing enhancing the potential for groundwater recharge. Aspect or slope direction in the STAS area ranged from -1 to 360 degrees. In groundwater studies, aspect indicates slope orientation, which affects solar exposure and in turn influences evaporation rates. The drainage buffer in the study area ranged from 0 to 9,695.46 m (Fig. 3e). It was classified as a continuous spatial variable to preserve the full variability of distances from drainage lines, allowing the models to capture subtle influences on groundwater potential. Spatially, zones closest to drainage features are

concentrated in the central part of the basin, whereas the most distal buffer zones occur predominantly in the western and eastern sections. Areas near drainage lines demonstrate improved interactions between surface and subsurface processes, and higher infiltration rates (Sockness and Gran 2022). These zones are more conducive to groundwater recharge than more distant areas. Drainage density in the study area ranged from 0.01 to $0.538 \text{ km per km}^2$, with the highest values concentrated in the central region. High drainage density indicates increased surface runoff and reduced infiltration, while low drainage density areas generally support higher infiltration and offer better groundwater potential (Ajayi et al. 2022; Ojo et al. 2024).

Geological and environmental data for groundwater predictive mapping is given in Fig. 4. Distances to lineaments range from 0 to 76,883.5 m. Zones close to lineaments were concentrated in the southern, central, and northern parts, while areas farther away were mostly in the north-eastern, eastern, and western sections. Areas near lineaments typically exhibit increased permeability, which enhances both the presence and movement of groundwater laterally and vertically. Consequently, these zones proximal to lineaments are likely to have high groundwater potential. Lineament density in the study area ranged from 0 to $0.126 \text{ km per km}^2$, with high-density zones mainly found in the central, northern, and southern regions. These areas of high lineament density indicate better groundwater potential, as they represent increased secondary porosity and permeability, which enhance groundwater storage and flow. Soil bulk density in the study area ranged from 1,368.59 to $1,557.48 \text{ kg/m}^3$, with the south-western and northern regions exhibiting higher values. Bulk density reflects soil compaction, where higher bulk density indicates poor groundwater potential because it reduces pore space, decreasing water infiltration and retention. In contrast, lower bulk density improves infiltration and supports groundwater recharge. Rainfall in the study area ranged from 214.11 to 571.26 mm, with higher rainfall concentrated along the northern part of the region. Rainfall is the main source of groundwater recharge. Increased rainfall enhances infiltration and aquifer replenishment, so areas receiving more rainfall generally have greater groundwater reserves, assuming infiltration is not limited. Soil wetness in the study area ranged from 0 to 255 DN, with the western half exhibiting higher wetness levels. Soil wetness measures how much water the soil's unsaturated (vadose) zone retains. It plays a crucial role by controlling the amount of precipitation that infiltrates the ground to recharge groundwater; higher soil wetness generally improves conditions for groundwater replenishment.

5 Groundwater Modelling Approaches

This study develops the first groundwater potential map for the Namibian portion of STAS by integrating GIS-based spatial analysis with advanced machine learning techniques. The modelling framework employs a stacking ensemble architecture, in which multiple base learners operate at the lower tier and a meta-learner consolidates their outputs at the upper tier. This hierarchical structure, widely recognised for improving predictive performance and model robustness (Dey and Mathur 2023), provides a suitable foundation for describing groundwater potential in a hydrogeological complex, data-limited environment.

The integration of machine learning with geospatial analysis facilitates a more accurate delineation of recharge zones, a better representation of groundwater flow patterns, and the identification of areas particularly susceptible to contamination (Bamal et al. 2024). This combined methodological approach improves our understanding of how the basin's hydrogeological systems work. It offers a practical decision support tool to inform sustainable groundwater management, particularly in light of increasing pressures from agricultural expansion and uranium exploration.

Within this overarching framework, the study adopts a structured, multi-stage procedure to develop the machine learning-based groundwater prediction models. Target data points were used to extract pixel values from the relevant exploration datasets using the extraction tool in ArcMap 10.3. These extracted values were compiled in Microsoft Excel and subsequently imported into Google Colab for pre-processing. The pre-processing workflow involved min–max normalisation of all predictor variables, followed by partitioning the dataset into an 80:20 train–test split to ensure sufficient data for robust model training. The 20% test subset was retained as an independent hold-out set for

unbiased final performance evaluation. Spatial cross-validation was applied during training to control spatial autocorrelation and assess model generalisation. Two machine learning algorithms were then trained on the processed data to generate spatial predictions of groundwater potential across the study area.

5.1 Boosted Regression Tree

BRT, also known as Gradient Boosted Trees, are a robust machine learning technique that combines regression trees with an iterative boosting algorithm to enhance predictive accuracy (Elith et al. 2008; Naghibi et al. 2016). The BRT models integrate multiple weak learners typically derived from Classification and Regression Trees in a sequential process, where each successive tree corrects the residual errors of its predecessor (Friedman 2001; Hastie et al. 2009). This ensemble strategy, similar to model averaging, aggregates outputs from many trees to improve overall model performance. Key tuning parameters in BRT include the learning rate, which controls the contribution of each tree. It also includes tree complexity, which defines the depth of variable interactions; and bag fraction, which determines the proportion of training data used at each iteration to introduce stochasticity (Elith et al. 2008; Rong et al. 2020). These components collectively enable BRT to model complex relationships with high accuracy and flexibility.

Equations (1–6) describe the statistical formulation of the BRT model. The model is initialised with a constant value that minimises the loss function across all training samples, such that:

$$F_0 = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^N L(y_i, \gamma) \quad (1)$$

Where y_i is observed target, and L is loss function.

Determine the negative gradient of the loss function relative to the predictions of the current model $F_{m-1}(x)$ at stage indices m :

$$r_{i,m} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)} \quad (2)$$

Construct a regression tree $h_m(x)$ using the residuals $r_{i,m}$ as the target variable, partitioning the input space into J terminal regions $R_{j,m}$, for $j = 1, 2, \dots, J$.

Here $h_m(x)$ is set to utilise the $r_{i,m}$ as the target variable. $r_{i,m}$ is the negative gradient of the i th sample in the m -th as the number of trees, $R_{j,m}$ represents the leaf node regions and j signifies leaf indices.

Table 2 Base and Meta learners, including key parameters, to enhance reproducibility of the SERF model implementation

Level	Learner Type	Key Parameters/Settings	Notes
Level 0 (Base Learner 1)	Decision Tree Classifier	Random states=2	Standard Decision Tree for classification
Level 0 (Base Learner 2)	Support Vector Machine	Kernel=radial basis function, Random states=2	Captures nonlinear patterns
Level 0 (Base Learner 3)	Gaussian Naive Bayes	Variance smoothing=1e-9, Default initialization, No prior	Probabilistic classifier
Level 1 (Meta Learner)	Logistic Regression	Random states=2	Combines base learner predictions into final output

Determine the leaf values $\gamma_{j,m}$ by minimising the loss function within each terminal region:

$$\gamma_{j,m} = \underset{\gamma}{\operatorname{argmin}} \sum_{x_i \in R_{j,m}} L(y_i, F_{m-1}(x_i) + \gamma) \quad (3)$$

Iteratively update the model by including the weighted contributions of new trees:

$$F_m(x) = F_{m-1}(x) + v \sum_{j=1}^J \gamma_{j,m} 1(x \in R_{j,m}) \quad (4)$$

Where $v \in (0,1]$ signifies the learning rate, regulating the impact of each tree (Natekin and Knoll 2013). This iterative method incrementally constructs the regression tree, with each subsequent tree progressively rectifying the residual mistakes of its predecessor to improve predictive accuracy.

Upon the conclusion of iterations, the final model is represented by Eq. 5. This iterative method progressively fits regression trees using the defects from each additional tree to inform the adjustments of succeeding trees, hence improving predicting accuracy.

$$F_M(x) = \sum_{m=1}^M v \sum_{j=1}^J \gamma_{j,m} 1(x \in R_{j,m}) \quad (5)$$

Moreover, the squared error $L(y, F(x)) = \frac{1}{2}(y - F(x))^2$ serves as the loss function for the deterministic estimate. In quantile regression, the loss function is delineated for a quantile $\alpha \in [0,1]$ (Elith et al. 2008; Natekin and Knoll 2013) as:

$$L(y_i, F(x_i), \alpha) = \begin{cases} \alpha(y_i - F(x_i)) & \text{if } y_i \geq F(x_i) \\ (\alpha - 1)(y_i - F(x_i)) & \text{otherwise} \end{cases} \quad (6)$$

BRT is especially advantageous for groundwater mapping because of its adaptability in simulating complex nonlinear relationships among factors such as slope, geology, soil type, precipitation, and land use. They can autonomously detect and represent variable interactions and nonlinearities, which is essential in hydrogeology where relationships are rarely linear. Moreover, BRT models are robust against missing data and outliers, requiring minimal pre-processing. Model validation used cross-validation, whereas prediction uncertainty was evaluated using ensemble averaging (Elith et al. 2008; Naghibi et al. 2016). In this study, the BRT model outputs were discretised using the Jenks natural breaks classification method. This approach identifies natural groupings in groundwater potential values more effectively than quantile classification. It enhances spatial interpretability and ensure

that mapped classes reflect meaningful hydrogeological differences (Jenks 1967).

5.2 Stacked Ensemble Random Forest

The SERF model is a sophisticated machine learning method designed to improve prediction accuracy by hierarchically integrating multiple base models (Liu et al. 2024; Mallick et al. 2022). This approach is particularly suitable for mapping groundwater potential in the STAS, where large spatial variability and complex hydrogeological conditions present analytical challenges. The SERF implementation follows a three-phase workflow given in Eq. 7 to 11.

- a) Level 0 Base Learner Training: Let $L_1, L_2, \dots, L_N$ denote N base Random Forest models on the same dataset provided by:

$$D = \{(x_i, y_i)\}_{i=1}^N \quad (7)$$

Where x_i represents the feature vector, and y_i denotes the matching class label.

Each base learner may employ distinct hyperparameters, feature subsets, or bootstrap samples to capture diverse data characteristics, producing independent predictions for both the training and testing datasets:

$$\hat{y}_i^{(k)} = L_k(x_i), k = 1, 2, \dots, N \quad (8)$$

Here $\hat{y}_i^{(k)}$ signifies the projected output of the k -th base learner for sample i ; defined as the continuous estimates for regression.

- b) At the meta level (Level 1), a new dataset is constructed using the predictions generated by all base learners, as defined by the following equation:

$$D_{meta} = \left\{ \left(\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \dots, \hat{y}_i^{(N)}; y_i \right) \right\}_{i=1}^N \quad (9)$$

A Meta learner is trained on D_{meta} to ascertain the most effective combination of base learner predictions, such that:

$$\hat{y}_i = f_{meta} \left(\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \dots, \hat{y}_i^{(N)} \right) \quad (10)$$

Where f_{meta} denotes the meta-classifier function.

For the unobserved data x_{new} , the SERF model generates final predictions given by:

$$\hat{y}_{new} = f_{meta} (L_1(x_{new}), L_2(x_{new}), \dots, L_N(x_{new})) \quad (11)$$

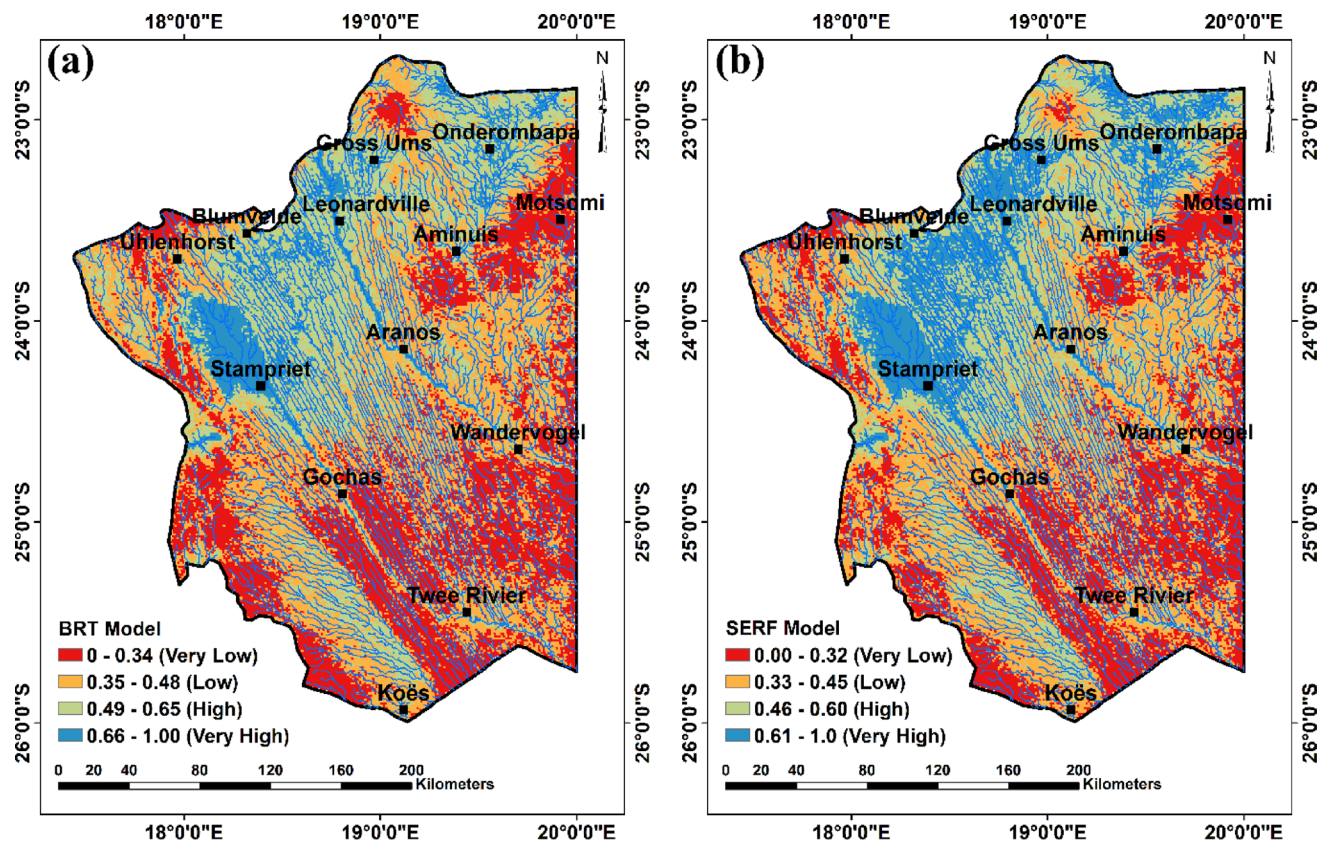


Fig. 5 Comparison of groundwater potential maps generated by the BRT and SERF models for the Stampriet Aquifer Basin in Namibia

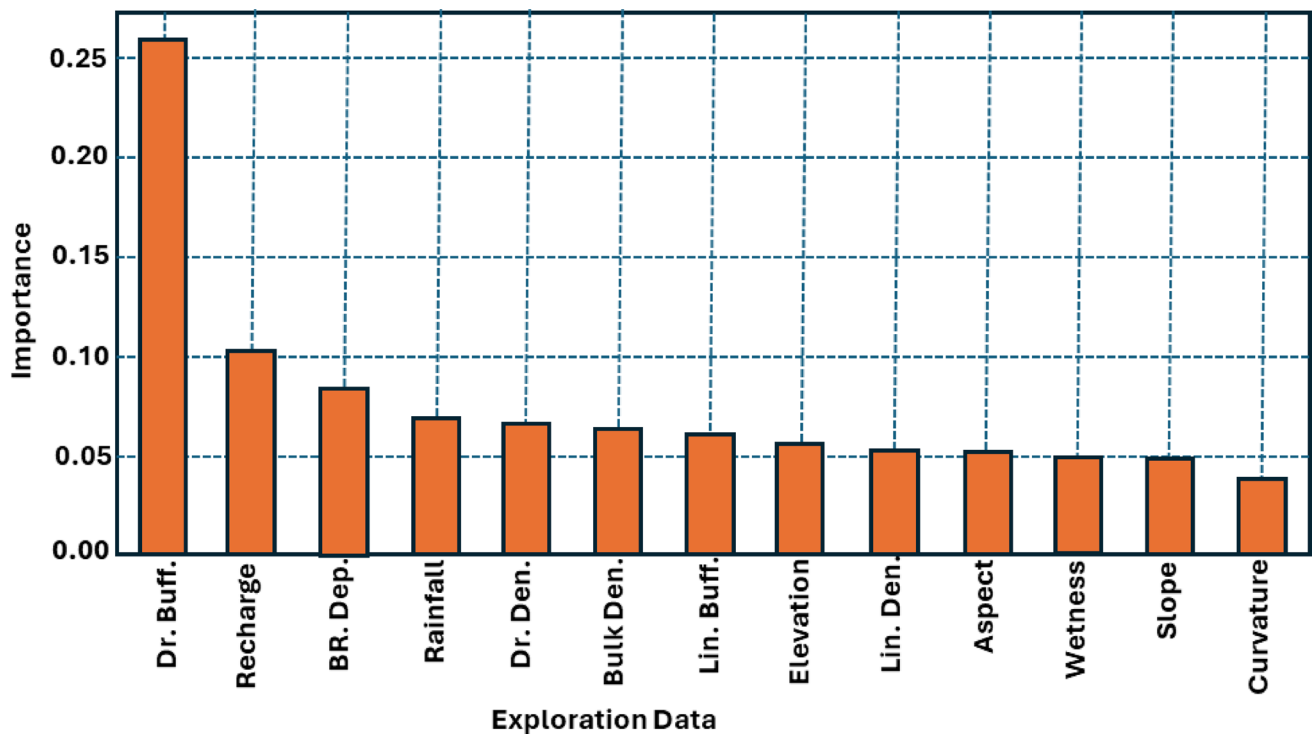


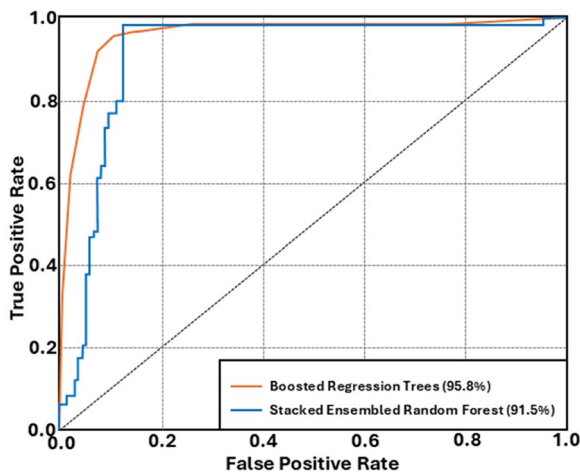
Fig. 6 Gini index purity plot showing feature importance from BRT classifier. Features include drainage buffer, recharge, bedrock depth, rainfall, drainage density, bulk density, lineament buffer, elevation, and lineament density

Table 3 Spatial distribution of groundwater potential classes derived from BRT and SERF model using natural breaks (Jenks) classification. Four classes are shown: very low, low, high, and very high

Sn	Prospectivity	BRT Model		SERF Model	
		Area (km ²)	Percentage (%)	Area (km ²)	Percentage
1.00	Very Low	17,938.54	28.36	16,022.03	25.33
2.00	Low	24,338.71	38.48	22,918.64	36.23
3.00	High	16,497.37	26.08	16,697.51	26.40
4.00	Very High	4,479.48	7.08	7,615.91	12.04

Table 4 Frequency ratios of groundwater potential classes for Boosted Regression Tree and Stacked Ensemble Random Forest models

Sn	Prospectivity	BRT	SERF
1	Very low	0.46	0.48
2	Low	0.54	0.46
3	High	1.20	1.12
4	Very high	4.92	3.47

**Fig. 7** ROC curves of the BRT and SERF models, with AUC values indicating predictive accuracy

Therefore, to enhance reproducibility, the Level 0 base learners consist of three classifiers capturing complementary data patterns: (i) a Decision Tree classifier with two random states, (ii) a Support Vector Machine with a radial basis function (RBF) kernel and two random states, and (iii) a Gaussian Naive Bayes classifier with default initialisation, no prior, and variance smoothing of 10^{-9} . At Level 1, a logistic regression Meta learner with two random states integrates the base learner outputs to produce the final groundwater potential prediction (Table 2; Liu et al. 2024; Mallick et al. 2022). This explicit specification ensures the SERF implementation is fully reproducible and transparent, facilitating replication or extension in future studies.

This configuration enables the ensemble to capture non-linear interactions, reduce overfitting, and improve generalisation relative to any single model, while remaining fully reproducible for future applications. These methodological advantages, including enhanced predictive accuracy, robustness to noisy datasets, and adaptability to heterogeneous

hydrogeological settings, make SERF well suited for regional-scale groundwater potential mapping.

6 Accuracy Assessment for Machine Learning Models

The prediction accuracies of the machine learning models were evaluated using ROC/AUC metrics. The ROC curve graphically illustrates the trade-off between the false positive rate (FPR) and true positive rate (TPR), showing how effectively a binary classification model distinguishes between classes at various threshold settings (McClish 1989). The AUC quantifies the overall performance of a diagnostic test by representing the average sensitivity across all possible specificity levels (Obuchowski 2003). The AUC values range from 0 to 1, where a value of 1 indicates perfect classification accuracy, and 0.5 corresponds to performance equivalent to random guessing (Fawcett 2001; Huang and Ling 2005). The test datasets facilitate the estimation of prediction accuracies for the different machine learning models. The TPR and FPR, which are necessary for constructing the ROC curve, are computed as given in Eqs. 12–13 below:

$$TPR = \text{Sensitivity} / \text{Recall} = TP / (TP + FN) \quad (12)$$

$$FPR = FP / (FP + TN) \quad (13)$$

where TP, FN, FP, and TN denote true positive, false negative, false positive, and true negative observations, respectively.

The AUC is then calculated statistically as:

$$AUC = \sum_{i=1}^{n+1} \frac{1}{2} \sqrt{(x_i - x_{i+1})^2} \cdot (y_i + y_{i+1}) \quad (14)$$

This procedure provides a robust, quantitative assessment of each model's ability to discriminate between areas of high and low groundwater potential.

6.1 Hyperparameters and Tuning

The study optimised model hyperparameters using five-fold spatial cross-validation combined with random search. For the BRT model, the tuning parameters included the learning rate (0.01, 0.05), tree depth (3, 5, and 7), number of estimators (300–1,000), and bag fraction (0.5, 0.7). In the SERF model, Level 0 base learners included Decision Trees with depths ranging from 3 to 15, and Support Vector Machines with radial basis function kernels. The SVM hyperparameters were tuned with a penalty parameter (C) from 1 to 10 and a kernel coefficient (γ) from 10^{-3} to 10^{-1} (Mammone et al. 2009). While ensemble level hyperparameters, including the number of estimators and bag fraction, were tuned over ranges of 300 to 10,000.

6.2 Frequency Ratio Analysis

The study computed the frequency ratio for each predictor class (k) using Eq. 15:

$$FR_k = \frac{N_k/N_{all}}{A_k/A_{all}} \quad (15)$$

where N_k represents the number of wells in class k and A_k is its corresponding area. A frequency ratio greater than 1 indicates a positive association with groundwater occurrence. To quantify uncertainty, 95% confidence intervals (CIs) for all frequency ratio were estimated via bootstrap resampling with $B=1,000$ iterations.

6.3 Receiver Operating Curve and Precision-Recall Metrics and Uncertainty

Model performance was assessed using multiple metrics, including the ROC-AUC, PR-AUC, balanced accuracy, and F1-score, all of which are robust to class imbalance. To quantify uncertainty, stratified bootstrap resampling ($B=1,000$) was applied at the cross-validation fold level, and 95% confidence intervals were reported for each metric.

6.4 Feature Importance and Interpretability

Variable importance and model interpretability were evaluated using permutation importance and SHAP values. These methods minimise Gini bias and assess predictor marginal effects through partial dependence analysis. The analysis

focused on the most influential predictors, including drainage buffer, recharge, bedrock depth, and rainfall.

7 Results

7.1 Application of BRT and SERF Machine Learning Models

The BRT and SERF models were applied to generate predictive maps of groundwater across the Stampriet Basin (Fig. 5). Both models consistently indicate that high-potential groundwater zones are concentrated from Stampriet northeast toward the Gross Ums and Onderombapa localities. In contrast, low-potential zones occur primarily in the central, southern, and eastern parts of the study area, including Gochas, Twee Rivier, Wandervogel, and Motsomi. BRT-based predictions range from 0.3 to 1, with a mean of 0.43 and a standard deviation of 0.14, providing a quantitative overview of groundwater potential across the basin. SERF-based predictions range from 0.00 to 1.00, with a mean of 0.42 and a standard deviation of 0.14.

A Gini index purity plot (Fig. 6) quantified the relative importance of predictor variables, identifying drainage buffer, recharge, and rainfall identified as the dominant controls on groundwater potential, with importance values of 0.258, 0.102 and 0.082, respectively. Wetness, slope and curvature were the least influential predictors, with low importance values of 0.0503, 0.0478, and 0.037 respectively.

To provide a clearer interpretation of groundwater potential, the continuous model outputs were discretised using the Jenks natural breaks classification. The resulting spatial distribution of groundwater potential classes is summarised in Table 3. For the BRT model, four classes were identified: the very low potential class (0.003–0.344) covers approximately 17,938.51 km² (28.36% of the study area), while the very high potential class (0.65–1.00) spans 4,479.47 km² (7.08% of the area). Similarly, the SERF model outputs were classified into four groundwater potential classes. The very low potential class (0.00–0.31) occupies 16,022.03 km² (25.33% of the study area), whereas the very high potential class (0.59–1.00) extends over 7,615.91 km² (12.04% of the total area). These classifications facilitate a clear comparison of high- and low potential zones across the basin and highlight subtle differences between the BRT and SERF model predictions.

Therefore, both the BRF and SERF models display a considerable degree of agreement in the spatial distribution of groundwater potential. However, a closer comparison indicates that very high groundwater potential zones are more prominently delineated in the SERF model than in the BRT model.

7.2 Validation of Groundwater Models

Visual interpretation of the groundwater prediction models revealed a strong spatial correlation between sustainable groundwater wells and areas classified as having high groundwater potential. This pattern was particularly evident in regions surrounding Stampriet in the western portion of the study area and Leonardville in the north.

The frequency ratio analysis supported this observation. For the BRT model, frequency ratios ranged from 0.46 in the very low groundwater potential zones to 4.92 in the very high potential zones. Similarly, the SERF model exhibited frequency ratios ranging from 0.48 in very low potential areas to 3.47 in the very high potential zones (Table 4).

The ROC/AUC curves (Fig. 7) further supported model performance, with curves closely aligned toward the upper-left quadrant, indicating strong predictive accuracy.

The computed AUC values demonstrated that the BRF model achieved a prediction accuracy of 95.8%, while the SERF model attained 91.5%, highlighting the slightly superior performance of the BRT approach in this study. Overall, these validation results confirm that both models are reliable tools for delineating groundwater potential zones, with the BRT model showing marginally higher accuracy in predicting high-potential areas.

8 Discussion

The groundwater potential maps derived from the BRT and SERF models reveal coherent spatial patterns. High to very high potential zones concentrate in the north-central, northern, and north-western STAS, extending from Stampriet into Gross Ums and Onderombapa. These zones cover small portions of the basin (7.08% for BRT and 12.04% for SERF), reflecting the limited recharge typical of arid, multi-layered aquifer systems. This spatial pattern aligns with hydrogeological understanding of the STAS, where groundwater occurrence depends on geological structure, stratigraphy, and focused recharge rather than uniform diffuse infiltration.

Although the groundwater potential maps appear as continuous surfaces, they capture the multi-layered nature of the STAS using predictor variables such as depth to bedrock, aquifer thickness, and structural connectivity between the Auob, Kalahari, and Nossob aquifers. This enables the models to identify areas of hydraulic connectivity or separation, providing a process-based understanding of recharge, storage, and flow. High potential zones coincide with hydrogeological favourable areas increasingly exposed to anthropogenic pressures, including uranium exploration. Limited recharge and aquifer connectivity in these zones indicate

heightened sensitivity to contamination. Although quantitative risk assessment was beyond this study's scope, the maps provide a spatial framework to prioritise aquifer protection, guide abstraction planning, and support transboundary groundwater governance.

Geological controls exert a primary influence on groundwater potential in the STAS. The dominance of Kalahari Group sands overlying the fractured Auob and Nossob sandstone aquifers creates a vertically stratified system. Permeability contrasts within this system regulate groundwater storage and flow. Machine learning models identify high potential zones in areas with shallow to moderate bedrock depth. These conditions enhance storage in unconsolidated sediments and improved hydraulic connectivity between shallow and deeper aquifers. This observation aligns with previous studies highlighting the importance of sediment thickness and bedrock configuration as key factors controlling groundwater availability in arid sedimentary basins (Lohe et al. 2020; Pedrazas et al. 2021).

Structural features further influence groundwater potential. Proximity to lineaments emerged as a secondary but significant control, reflecting the role of faults and fractures in enhancing secondary porosity and transmissivity. Similar patterns occur in arid regions of Saudi Arabia and Egypt, where high groundwater potential coincides with areas of elevated lineament density and fractured sedimentary or basement rocks (Khan and ElKashouty 2023; Khan et al. 2023). In the STAS, the influence of structural features is spatially variable. Some faults act as preferential flow paths, while others impede flow due to clay-rich fault gouge. This dual role explains why lineament related variables ranked below drainage and recharge in overall model importance.

Hydrochemical and hydrological factors, particularly recharge along ephemeral drainage networks, emerged as dominant controls on groundwater potential. The drainage buffer variable ranked highest in the BRT model, highlighting ephemeral rivers and palaeochannels as key infiltration pathways in a recharge limited environment. This finding supports previous isotopic and conceptual studies of the Stampriet Basin, which identified transmission losses from ephemeral streams as the primary recharge mechanism for shallow and semi-confined aquifers (Kirchner et al. 2002; Stone and Edmunds 2012). Similar patterns occur in arid basins globally. In the Tabuk Basin Saudi Arabia, areas near drainage networks with permeable sediments consistently exhibit higher aquifer potential, even under low annual rainfall conditions (Khan et al. 2023).

Although rainfall represents the ultimate source of recharge, it ranked lower than drainage-related variables in predictive importance. This reflects the dominance of high evapotranspiration rates and the episodic nature of precipitation in the STAS. Only a small fraction of rainfall

contributes to groundwater, primarily during high intensity events that generate surface runoff capable of infiltrating along drainage corridors. Similar patterns occur in arid regions of China and the Middle East, where rainfall alone shows weak correlation with groundwater occurrences unless combined with favourable geomorphological and geological conditions (Li et al. 2024; Shanafield and Cook 2014). These results indicate that groundwater potential in the STAS is controlled less by total rainfall and more by the spatial and process-based redistribution of water into recharge zones.

Topographic variables, including slope, curvature, and aspect, exerted comparatively low influence on groundwater potential. This reflects the generally flat terrain of the STAS, where gradients weakly affect infiltration and runoff at the basin scale. In contrast, studies in more rugged arid regions, such as Aseer–Jazan (Saudi Arabia), show stronger topographic-groundwater relationships due to enhanced runoff concentration and orographic effects (Khan et al. 2022a). These findings highlight the need to tailor groundwater potential models to basin-specific geomorphic contexts.

Model performance further emphasises the robustness of the BRT approach, which achieved an AUC of 0.958, compared to 0.915 for the SERF model. This superior performance indicates that the BRT captures nonlinear interactions between geological and hydrological controls more effectively. This is consistent with previous studies showing boosted tree-based models outperforming random forest and stacked ensembles by resolving threshold effects and complex variable interaction (Naghibi et al. 2016; Sachdeva and Kumar 2021). Importantly, the spatial alignment of high potential zones with productive boreholes supports model realism, as reflected by high frequency ratio values exceeding 4.9 for BRT. Comparable validation results occur in other arid basins, where machine learning predictions align with borehole yields and favourable hydrochemical indicators including low TDS and nitrate concentrations (Khan et al. 2022a; Rahim et al. 2025). The identification of spatially discrete high potential zones has important implications for groundwater management in the STAS. These zones coincide with hydrogeological favourable areas increasingly exposed to anthropogenic pressures, including irrigation expansion and uranium exploration. Given the multi-layered aquifer system and limited recharge, indiscriminate abstraction outside high potential zones may accelerate groundwater level declines. Such practices risk degrading artesian conditions, particularly in the Auob aquifer.

High groundwater potential and primary recharge zones in the north-central and north-western STAS show heightened sensitivity to uranium in situ leaching (ISL). The models identify hydrogeological favourable areas where recharge corridors, aquifer connectivity, and structural

features coincide with ongoing or planned uranium exploration. Faults, erosional windows, and weakened confining layers above the Auob Aquifer may enable vertical migration of leachates into the Kalahari (Mwetulundila and Atangana 2025). Recharge corridors along ephemeral rivers may further facilitate contaminant transport, even where surface indicators suggest low risk. Although, machine learning maps support regional screening, they rely on proxy variables and cannot fully resolve subsurface processes. Consequently, site-specific investigations, including fault mapping, confinement assessments, and reactive transport modelling, remain essential to protect recharge zones and support evidence-based groundwater management.

Building on these insights, this study established a scientifically robust framework for zoning groundwater abstraction, safeguarding recharge corridors, and informing transboundary land-use planning. The framework explicitly links groundwater potential patterns to underlying geological and hydrological processes. Similar groundwater potential mapping approaches have effectively supported aquifer protection and sustainable development in other arid transboundary settings (Khan et al. 2023; Mallick et al. 2022). Overall, the study shows that process-informed ensemble machine learning can meaningfully support groundwater potential assessment in arid transboundary aquifers when applied alongside hydrogeological understanding.

9 Limitations of the study

Although the machine learning models achieved high predictive efficiencies (>90%), these values represent relative discrimination ability rather than absolute certainty. In a data-limited, multi-layered arid aquifer system such as the STAS, residual spatial autocorrelation may still influence model performance. The use of pseudo-absence samples may include unidentified groundwater occurrences, potentially biasing model calibration. Reliance on surface-derived proxy variables further constrains the ability to resolve subsurface hydrogeological complexity. In addition, predictor layers varied in spatial resolution and temporal coverage, which may affect internal model consistency. Structural features were extracted through expert-guided manual digitisation of aeromagnetic data. Although this approach improves geological relevance, it introduces subjectivity and interpreter bias that may influence lineament density and proximity variables. The BRT model may also exhibit overfitting if learning rate and tree depth are insufficiently constrained. In addition, the transferability of model outputs beyond the current climatic conditions and existing anthropogenic pressures remains uncertain.

Accordingly, the mapped groundwater potential zones should be interpreted as probabilistic indicators of groundwater favourability rather than definitive estimates of aquifer productivity. These outputs are most suitable for regional screening and prioritising field investigations. Future research should integrate subsurface information including pump-test yields, borehole lithology, and hydrochemical or isotopic data, to improve model calibration. Applying stricter spatial validation approaches such as leave-one-region-out cross-validation, would further reduce spatial bias. Explicit representation of inter-aquifer interactions and adoption of reproducible modelling workflows would enhance robustness, transferability, and long-term applicability of the results.

10 Conclusion

This study developed the first machine learning-derived groundwater potential map for the Namibian portion of STAS. The results demonstrate that ensemble tree-based methods can generate operationally useful groundwater prospectivity outputs in a hyper-arid, data-limited, multi-layered aquifer setting. Using spatial block cross-validation and complementary validation metrics, including frequency ratios and ROC–AUC, both BRT and SERF produced coherent and interpretable groundwater potential patterns. The BRT model achieved stronger discrimination ($AUC=0.958$) than SERF ($AUC=0.915$), indicating improved sensitivity to nonlinear hydrogeological controls. Key predictors, including drainage proximity, net recharge, rainfall, and depth to bedrock, indicate that groundwater occurrence in STAS is governed primarily by focused infiltration along ephemeral drainage pathways. Subsurface storage controls further regulate groundwater availability, rather than rainfall amount alone. High to very high groundwater potential zones concentrate within a strategic north-central to north-western corridor, occupying approximately 7.08–12.04% of the mapped area. This spatial clustering indicated that productive groundwater opportunities are limited and unevenly distributed across the basin.

These findings provide a defensible evidence base for prioritizing exploratory drilling, improving borehole siting efficiency, and strengthening groundwater licensing and protection zoning. Such guidance is increasingly important under rising abstraction pressures and expanding land-use risks within the STAS. The delineated high potential and primary recharge zones also supported transboundary groundwater governance and identified areas requiring careful management. This is particularly relevant for mitigating risks associated with uranium in situ leaching, intensive

abstraction, and other anthropogenic pressures on shared aquifer resources.

Future research should integrate subsurface and monitoring data, including pump-test yields, borehole lithology, hydrochemistry, and isotopic information, to improve model calibration. Applying stricter spatial validation strategies, such as leave-one-region-out cross-validation, would further reduce spatial bias and uncertainty. Operationalising reproducible workflows will enable periodic updates as new data become available. Careful attention to input data resolution and consistency will further improve predictive accuracy and the reliability of groundwater potential delineations. Field verification and protection of high potential and primary recharge zones remain essential to support sustainable groundwater management and informed land-use planning.

Acknowledgements The authors appreciate the discussions and assistance from Dr Tende Andongma and the University of the Free State for providing access to the instruments and software used in this analysis.

Authors Contribution Both authors contributed equally to the development and completion of this manuscript.

Funding Open access funding provided by University of the Free State. The University of the Free State supported the open access. No other funding or grant were provided during this research study.

Data Availability The data are available on request from the corresponding authors.

Declarations

Competing Interests Both authors declare there are no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Adeoye AS, Ajayi CA, Ibrahim OI (2025) Application of integrated geophysical methods for groundwater exploration in Iworoko-Ekiti, Southwestern Nigeria. *FNAS-JAPS* 2(2):28–38. <https://fnajournals.com/index.php/FNAS-JAPS/article/view/734>

- Ahaneku CV, Obiamalu CC, Aseh P, Elomba UF, Muogbo CD, Ogbuefi CE (2025) Integration of geophysical methods for comprehensive site characterization: Principles and case studies. *Int J Sci Res Arch* 03:1010–1025. <https://doi.org/10.30574/ijrsra.2025.15.3.1788>
- Ajayi OG, Nwadiakor IJ, Odumosu JO et al (2022) Assessment and delineation of groundwater potential zones using integrated geospatial techniques and analytic hierarchy process. *Appl Water Sci* 12:276. <https://doi.org/10.1007/s13201-022-01802-4>
- Alao JO, Abubakar F (2025) Groundwater exploration, management strategies and sustainability: Geophysical approaches. *GeoGeo* 4(3):100395. <https://doi.org/10.1016/j.geogeo.2025.100395>
- Ali S, Liu D, Fu Q, Cheema MJM, Pham QB, Rahaman MM, Dang TD, Anh DT (2021) Improving the resolution of GRACE data for spatio-temporal groundwater storage assessment. *Remote Sens* 13(17):3513. <https://doi.org/10.3390/rs13173513>
- Bamal A, Uddin MG, Olbert AI (2024) Harnessing machine learning for assessing climate change influences on groundwater resources: a comprehensive review. *Heliyon*. <https://doi.org/10.1016/j.heliyon.2024.e37073>
- Calow RC, MacDonald AM, Nicol AL, Robins NS (2010) Groundwater security and drought in Africa: Linking availability, access, and demand. *Ground Water* 48(2):246–256. <https://doi.org/10.1111/j.1745-6584.2009.00558.x>
- Centre for Hydrometry and Remote Sensing (2025) CHRS iRain – An integrated system for global real-time precipitation observation using PUnet. <https://irain.eng.uci.edu>
- Cihan A, Zhou Q, Birkholzer JT, Kraemer SR (2014) Flow in horizontally anisotropic multi-layered aquifer systems with leaky wells and aquitards. *Water Resour Res* 50(1):741–747. <https://doi.org/10.1002/2013WR013867>
- Corner B, Durrheim RJ (2018) An integrated geophysical and geological interpretation of the southern African lithosphere. In: Siegesmund S, Basei M, Oyhançabal P, Oriolo S (eds) *Geology of Southwest Gondwana. Regional Geology Reviews*. Springer, pp 19–61
- Dey R, Mathur R (2023) Ensemble learning method using stacking with base learner, A comparison. In: Chaki N, Roy ND, Debnath P, Saeed K (eds), *Proceedings of International Conference on Data Analytics and Insights (ICDAI) 2023*. Lecture notes in networks and systems, 727. Springer, Singapore. https://doi.org/10.1007/978-981-99-3878-0_14
- Elith J, Leathwick JR, Hastie T (2008) A working guide to boosted regression trees. *J Anim Ecol* 77(4):802–813. <https://doi.org/10.1111/j.1365-2656.2008.01390.x>
- Fawcett T (2001) Using rule sets to maximize ROC performance. In *Proceedings 2001 IEEE international conference on data mining* 131–138. <https://doi.org/10.1109/ICDM.2001.989510>
- Friedman JH (2001) Greedy function approximation: a gradient boosting machine. *Ann Statist* 29(5):1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Golovina E, Grebneva A (2021) Some aspects of groundwater resources management in transboundary areas. *J Ecol Eng* 22(4):106–118. <https://doi.org/10.12911/22998993/134037>
- Haghirri M, Raeisi N, Azizi R, Shabani K, Ghadirri M (2024) Evaluation of karst aquifer development and karst water resource potential using fuzzy logic model (FAHP) and Analysis Hierarchy Process (AHP): a case study, North of Iran. *Carbonates Evaporites* 39:11. <https://doi.org/10.1007/s13146-024-00925-w>
- Halder K, Srivastava AK, Ghosh A, Nabik R, Pan S, Chatterjee U, Bisai D, Pal SC, Zeng W, Ewert F, Gaiser T, Pande CB, Islam ARMT, Alam E, Islam MK (2024) Application of bagging and boosting ensemble machine learning techniques for groundwater potential mapping in a drought-prone agriculture region of eastern India. *Environ Sci Eur* 36:155. <https://doi.org/10.1186/s12302-024-00981-y>
- Hastie T, Tibshirani R, Friedman J (2009) *The elements of statistical learning. Data mining, inference, and prediction*. Springer Series in Statistics, 2nd edn. Springer
- Huang J, Ling CX (2005) Using AUC and accuracy in evaluating learning algorithms. *IEEE Trans Knowl Data Eng* 17(3):299–310. <https://doi.org/10.1109/TKDE.2005.50>
- ISRIC (International Soil Reference and Information Centre) (2025) Data and resources. <https://isric.org/explore/>
- Japan International Cooperation Agency (2002) *The study on the groundwater potential evaluation and management plan in the southeast Kalahari (Stampriet) artesian basin in the Republic of Namibia: Final Report. Four Volumes* https://openjicareport.jica.go.jp/618/618/618_522_11681707.html
- Jenks GF (1967) The data model concept in statistical mapping. *Int Yearbook Cartography* 7:186–190
- Jet Propulsion Laboratory, University of California (2004) ASTER global digital elevation map announcement. <https://asterweb.jpl.nasa.gov/gdem.asp>
- Khan MYA, ElKashouty M (2023) Watershed prioritization and hydro-morphometric analysis for the potential development of Tabuk Basin, Saudi Arabia using multivariate statistical analysis and coupled RS-GIS approach. *Ecol Indic* 154:110766. <https://doi.org/10.1016/j.ecolind.2023.110766>
- Khan MYA, ElKashouty M, Subyani AM, Tian F, Gusti W (2022b) GIS and RS intelligence in delineating groundwater potential zones in arid southern Aseer, Saudi Arabia. *Appl Water Sci* 12:3. <https://doi.org/10.1007/s13201-021-01535-w>
- Khan MYA, ElKashouty M, Tian F (2022a) Mapping groundwater potential zones using AHP and MCE in Egypt. *Water* 14(7):1041. <https://doi.org/10.3390/w14071041>
- Khan MYA, ElKashouty M, Zaidi FK, Egbueri JC (2023) Mapping aquifer recharge potential zones using integrated geospatial AHP: a case from Saudi Arabia. *Remote Sens* 15:2567. <https://doi.org/10.3390/rs15102567>
- Kinoti I, Leblanc M, Lekula M, Tweed S, Kenabatho PK, Oliosio A, Lubczynski MW (2024) Hydrogeological conceptual model of Stampriet transboundary aquifer system in Southern Africa. *Groundwater for Sustainable Development* 26:101301. <https://doi.org/10.1016/j.gsd.2024.101301>
- Kirchner J, Tredoux G, Wierenga A, Christelis G, Wrabel J (2002) Namibia: assessment of the recharge to the stampriet artesian basin to formulate a groundwater management plan for sustainable use of the resource in the Southeast Kalahari in the Republic of Namibia - Technical Report. IAEA RAF/8/029, p 32
- Li G, Yao J (2024) A review of in situ leaching (ISL) for uranium mining. *Mining* 4(1):120–148. <https://doi.org/10.3390/mining4010009>
- Li M, Xie Y, Dong Y, Wang L, Zhang Z (2024) Recent progress on groundwater recharge research in arid and semiarid areas of China. *Hydrogeol J* 32(1):9–30. <https://doi.org/10.1007/s10040-023-02656-z>
- Liu J, Chen J, Chen S, Wu K (2024) Forest management type identification based on stacking ensemble learning. *Forests* 15(5):887. <https://doi.org/10.3390/f15050887>
- Li Z, Wang G, Wang X, Wan L, Shi Z, Wanke H, Uugulu S, Uahengo C-I (2018) Groundwater quality and associated hydrogeochemical processes in Northwest Namibia. *J Geochem Explor* 186:202–214. <https://doi.org/10.1016/j.gexplo.2017.12.015>
- Lohe C, Amster R, Swartz B (2020) *Groundwater in Namibia: An explanation of the hydrogeological map*. Ministry of Agriculture-Water and Rural Development. Windhoek, Namibia. https://www.bgr.bund.de/EN/Themen/Grundwasser/Downloads/groundwater_namibia_pdf.pdf
- Mallick J, Talukdar S, Ahmed M (2022) Combining high resolution input and stacking ensemble machine learning algorithms for developing robust groundwater potentiality models in Bisha

- watershed, Saudi Arabia. *Appl Water Sci* 12(4):77. <https://doi.org/10.1007/s13201-022-01599-2>
- Mammone A, Turchi M, Cristianini N (2009) Support vector machines. *Wiley Interdisciplinary Reviews: Comput Stat* 1(3):283–289. <http://doi.org/10.1002/wics.49>
- Mapani BS, Shikangalah RN, Mwetulundila AL (2023) A review on water security and management under climate change conditions, Windhoek, Namibia. *J Afri Earth Sci* 197:104749. <https://doi.org/10.1016/j.jafrearsci.2022.104749>
- Maqsoom A, Aslam B, Khalid N, Ullah F, Anysz H, Almaliki AH, Almaliki AA, Hussein EE (2022) Delineating groundwater recharge potential through remote sensing and geographical information systems. *Water* 14(11):1824. <https://doi.org/10.3390/w14111824>
- McClish DK (1989) Analyzing a portion of the ROC curve. *MDM* 9(3):190–195. <https://doi.org/10.1177/0272989X8900900307>
- Mekonnen MM, Hoekstra AY (2016) Four billion people facing severe water scarcity. *Sci Adv*. <https://doi.org/10.1126/sciadv.1500323>
- Mendelsohn J, Jarvis A, Roberts C, Robertson T (2002) Atlas of Namibia: A Portrait of Land and its People, vol 53. David Philip, South Africa
- Miller RM, Carr R, Van der Merwe B, Symons G, Murray R, Amputu M (2018) Some geological and geophysical aspects of the 2016/2017 drilling campaign in the Windhoek Aquifer. *Commun Geol Surv Namibia* 19:8–19. https://mme.gov.na/files/publications/688_GSN%20Comms%2019.%202020Miller%20et%20al_Windhoeck%20Aquifer.pdf
- Miller RMG (2008) The geology of Namibia: Neoproterozoic to lower palaeozoic. The Geology of Namibia, Geology Survey (Namibia), 2. Ministry of Mines and Energy, Geological Survey, 2008
- Miller RMG (2013) Comparative stratigraphic and geochronological evolution of the Northern Damara Supergroup in Namibia and the Katanga Supergroup in the Lufilian Arc of Central Africa. *Geosci Can* 40(2):118–140. <https://doi.org/10.12789/geocanj.2013.40.007>
- Miller RMG (2024) Environmental threats posed by the proposed in-situ leach mining of uranium to underground potable water aquifers in the Stampriet artesian basin. *Namib J Soc Just* 4:1–10
- Mudd GM (2022) Acid in situ leach uranium mining—1. USA and Australia. In: Tailings and Mine Waste 2000 edited by A.A. Balkema Publishers . 517–526. London: CRC Press. <https://doi.org/10.1201/9781003078579>
- Mwetulundila AL, Atangana A (2025) Numerical simulation of uranium in-situ leaching contaminant transport in a spatially heterogeneous confined aquifer. *Model Earth Syst Environ* 11(5):316. <https://doi.org/10.1007/s40808-025-02462-4>
- Naghbi SA, Pourghasemi HR, Dixon B (2016) GIS-based groundwater potential mapping using boosted regression tree, classification and regression tree, and random forest machine learning models in Iran. *Environ Monit Assess* 188:44. <https://doi.org/10.1007/s10661-015-5049-6>
- NASA (National Aeronautics and Space Administration). Earthdata. <https://www.earthdata.nasa.gov/>
- Natekin A, Knoll A (2013) Gradient boosting machines: a tutorial. *Front Neurobot* 7:21. <https://doi.org/10.3389/fnbot.2013.00021>
- Nijsten GJ, Christelis G, Villholth KG, Braune E, Gaye CB (2018) Transboundary aquifers of Africa: review of the current state of knowledge and progress towards sustainable development and management. *J Hydrol Reg Stud* 20:21–34. <https://doi.org/10.1016/j.ejrh.2018.03.004>
- Nuunyang EMN (2019) The influence of groundwater management strategies on groundwater levels of Namibian savannah aquifers (MSc thesis), University of Namibia
- Obuchowski NA (2003) Receiver operating characteristic curves and their use in radiology. *Radiology* 229(1):3–8. <https://doi.org/10.1148/radiol.2291010898>
- Odoh BI, Ahaneku CV, Madu FM, Muogbo CD, Anozie HC, Arukwe-Moses CP, Eze IE (2024) Groundwater exploration and management using geophysical and remote sensing techniques: a review. *Int J Sci and Manag Stud* 7(4):230–243. <https://doi.org/10.51386/25815946/ijms-v7i4p128>
- Ojo OT, Chiaka IJ, Nwokeabia CN (2024) Integrated geophysical and GIS approaches for groundwater potential assessment: a case study of Aladja, Delta State, Nigeria. *Water Pract Technol* 19(10):4282–4302. <https://doi.org/10.2166/wpt.2024.254>
- Olaleye OB (2024) Investigating the synergistic fusion of GIS and machine learning in multi-criteria analysis for groundwater potential assessment. *ResearchGate*. <https://doi.org/10.13140/RG.2.2.17711.24481>
- ORASECOM (2016) Stampriet transboundary aquifer system (STAS): Governance of groundwater resources in Transboundary Aquifers (GGRETA) project Phase 1. Technical Report, 171. <https://wis.orasecom.org/content/study/UNESCO/STAS/>
- Pappaka RK, Nakkala AB, Badapalli PK, Gugulothu S, Anguluri R, Hasher FFB, Zhran M (2025) Machine learning-driven groundwater potential zoning using geospatial analytics and random forest in the Pandameru river basin, South India. *Sustainability* 17(9):3851. <https://doi.org/10.3390/su17093851>
- Pawar NS, Sharma KV (2025) Advanced fuzzy-based landslide prediction in high-risk regions using hybrid and statistical GIS models. *Phys Geogr* 46(4):307–337. <https://doi.org/10.1080/02723646.2025.2542909>
- Pedrazas MA, Hahm WJ, Huang MH, Dralle D, Nelson MD, Breunig RE, Fauria KE, Bryk AB, Dietrich WE, Rempe DM (2021) The relationship between topography, bedrock weathering, and water storage across a sequence of ridges and valleys. *J Geophys Res Earth Surf* 126(4):e2020JF005848. <https://doi.org/10.1029/2020JF005848>
- Rahim OA, Yin H, Ullah S, Durrani AN (2025) Delineation of groundwater potential zones and recharge using multi-source big data and systematic analysis approach. *Sustain Futures* 10:100872. <https://doi.org/10.1016/j.sfsr.2025.100872>
- Rodriguez MMC, Ferolin TP (2024) Groundwater resource exploration and mapping methods: a review. *J Environ Eng Sci* 19(3):140–156. <https://doi.org/10.1680/jenes.23.00051>
- Rong G, Alu S, Li K, Su Y, Zhang J, Zhang Y, Li T (2020) Rainfall induced landslide susceptibility mapping based on Bayesian optimized random forest and gradient boosting decision tree models – a case study of Shuicheng County, China. *Water* 12(11):3066. <https://doi.org/10.3390/w12113066>
- Sachdeva S, Kumar B (2021) Comparison of gradient boosted decision trees and random forest for groundwater potential mapping in Dholpur (Rajasthan), India. *Stoch Environ Res Risk Assess* 35(2):287–306. <https://doi.org/10.1007/s00477-020-01891-0>
- Saleh MA, Rasel HM (2024) Machine learning for groundwater levels: uncovering the best predictors. *Sustain Water Resour Manag* 10(5):166. <https://doi.org/10.1007/s40899-024-01146-8>
- Seredkin M, Zabolotsky A, Jeffress G (2016) In situ recovery, an alternative to conventional methods of mining: exploration, resource estimation, environmental issues, project evaluation and economics. *Ore Geol Rev* 79:500–514. <https://doi.org/10.1016/j.joregeorev.2016.06.016>
- Shanfield M, Cook PG (2014) Transmission losses, infiltration and groundwater recharge through ephemeral and intermittent streambeds: a review of applied methods. *J Hydrol* 511:518–529. <https://doi.org/10.1016/j.jhydrol.2014.01.068>
- Shikangalah RN (2020) The 2019 drought in Namibia: an overview. *JNS & History, Politics, Culture* 27:37–58
- Shrestha N (2020) Detecting multicollinearity in regression analysis. *Am J Appl Math Stat* 8(2):39–42. <https://doi.org/10.12691/ajams-8-2-1>

- Sockness BG, Gran KB (2022) An experimental study of drainage network development by surface and subsurface flow in low-gradient landscapes. *Earth Surf Dyn* 10(3):581–603. <https://doi.org/10.5194/esurf-10-581-2022>
- Stone AEC, Edmunds WM (2012) Sand, salt and water in the Stampriet Basin, Namibia: calculating unsaturated zone (Kalahari dunefield) recharge using the chloride mass balance approach. *Water SA* 3:367–378. <https://doi.org/10.4314/wsa.v38i3.2>
- Tredoux G, Kirchner J, Miller RM, Yamasaki Y, Christeli GM, Wierenga A (2002) Redefining the recharge behaviour of the Stampriet Artesian Basin, Namibia. In: *Proceedings of the IAH Conference on Balancing the Groundwater Budget*. 12–17 May, 2002 (Darwin, Australia)
- UN-Water (2025) Water scarcity – Water facts. <https://www.unwater.org/water-facts/water-scarcity>
- Ugulu S, Wanke H (2020) Estimation of groundwater recharge in savannah aquifers along a precipitation gradient using chloride mass balance method and environmental isotopes, Namibia. *Physics and Chemistry of the Earth, Parts A/B/C* 116:102844. <https://doi.org/10.1016/j.pce.2020.102844>
- Xu X, Chen F, Wang B, Harrison MT, Chen Y, Liu K, Xhang C, Zhang X, Feng P, Hu K (2024) Unleashing the power of machine learning and remote sensing for robust seasonal drought monitoring: a stacking ensemble approach. *J Hydrol* 634:131102. <https://doi.org/10.1016/j.jhydrol.2024.131102>
- Yusuf M, Omolaiye GE, Oluyemoh AO, Ajadi J, Adam SB, Egbeyale GB, Alade AD (2025) Hydrogeological assessment of groundwater potential zones using remote sensing, geographical information system (GIS) and aeromagnetic data at Kwara State University, Malete, Nigeria. *Discov Water* 5(1):72. <https://doi.org/10.1007/s43832-025-00273-3>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.